

Kapture Finance — AI Collections Voicebot

AI Delivery Intern – Take-Home Assignment

Prepared by: Thanuja

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1. Objective

Kapture Finance requires an outbound AI voice agent that can contact customers with overdue loan EMIs and handle routine collections conversations politely, safely, and efficiently.

The proposed voice agent, **Maya**, will introduce itself, identify the customer, authenticate the customer before revealing account-specific information, explain the overdue payment, understand the customer's intent, and guide the customer toward an appropriate resolution.

Depending on the customer's response, Maya can record a promise to pay, provide a payment link, handle disputes or hardship situations, schedule a callback, respect do-not-call requests, or escalate the conversation to a human agent.

Every completed call will have a structured final disposition so that the outcome can be tracked and audited.

1.1 Example Customer

Field	Value
Customer	Rahul Sharma
Loan Type	Personal Loan
Overdue EMI	\$ 8,499
Days Past Due	12
Customer ID	RAHUL123

Note: The customer information above is mock data used only for the assignment demonstration.

2. System Architecture

The proposed collections voicebot, **Maya**, uses **Vapi** as the voice-agent platform. The system connects the customer's phone call with speech recognition, an AI conversation orchestrator, backend tools/APIs, and text-to-speech.

Overall Pipeline

Customer → Telephony → Speech-to-Text (STT) → LLM/Orchestrator → Tools/APIs → Text-to-Speech (TTS) → Customer

How the Pipeline Works

1. **Customer:** The customer receives an outbound call from Maya.
2. **Telephony:** The phone connection carries the customer's voice to the voicebot and returns Maya's response.
3. **Speech-to-Text (STT):** The customer's speech is converted into text so that the AI can understand the response.
4. **LLM / Conversation Orchestrator:** The LLM interprets the customer's response, identifies the customer's intent, and determines the next action according to the current conversation state.
5. **Tools / APIs:** When an action requires backend processing, Maya calls a specific tool/API. Examples include customer verification, logging a promise to pay, sending a payment link, escalating to a human agent, and recording the final call disposition.
6. **Text-to-Speech (TTS):** Maya's response is converted from text into natural speech and played back to the customer.

2.1 System Components

Component	Responsibility
Telephony	Connects Maya with the customer through an outbound phone call
Speech-to-Text(STT)	Converts customer's speech into text
Vapi	Manages the voice-agent

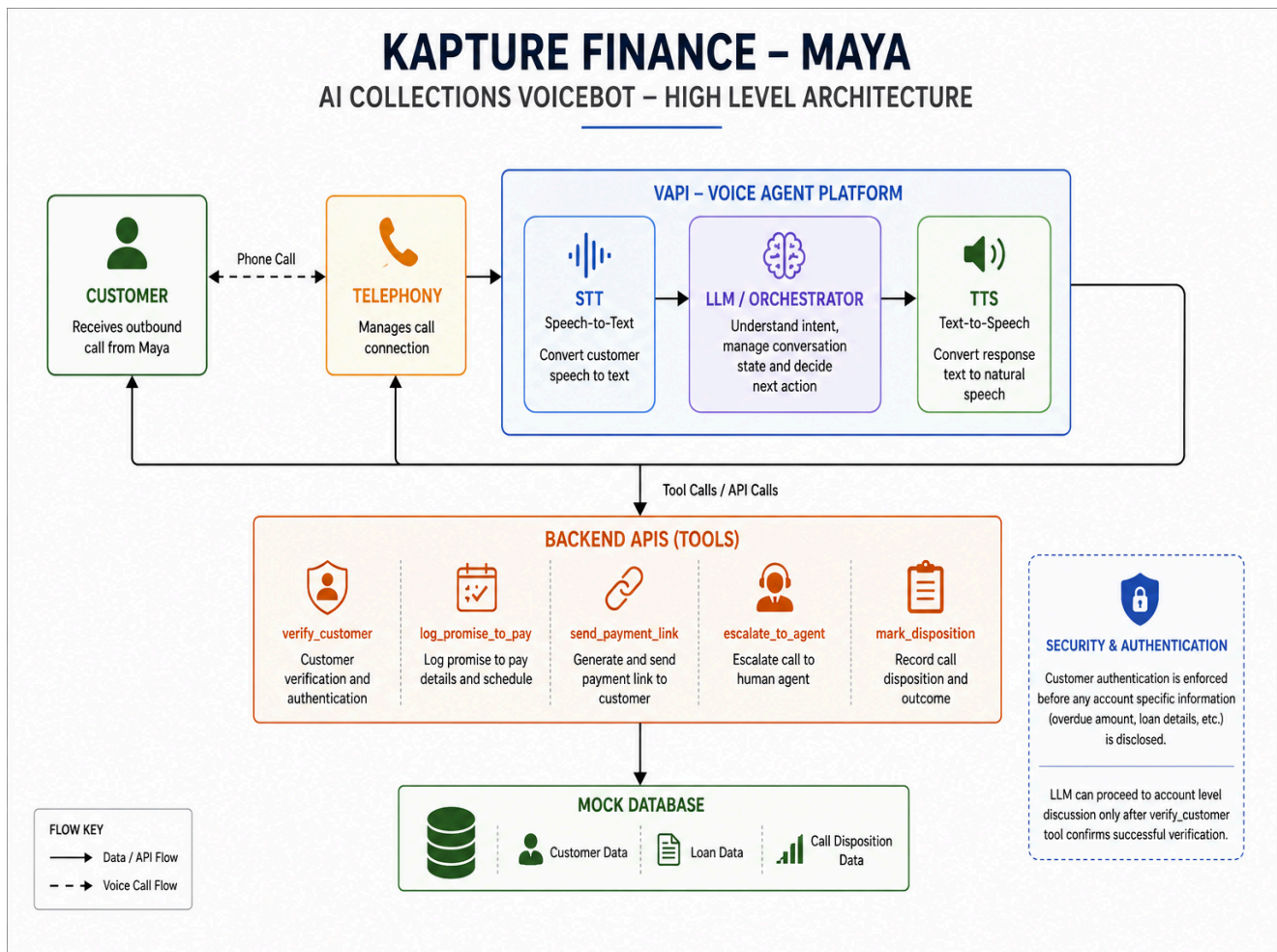
	conversation
LLM/Orchestrator	Understands customer intent and determines the next action
Backend APIs	Perform verification, PTP logging, payment-link, escalation and disposition operations
Text-to-speech(TTS)	Converts Maya's response into natural speech
Mock Database	Stores customer, loan and call-disposition data

2.2 Latency Targets

Component	Target Latency
Telephony / Network	100-300 ms
STT	200-500 ms
LLM Response	300-800 ms
Backend Tool/API	< 300 ms
TTS First Audio	200-500 ms
Target Turn Latency	< 1.5-2 seconds

These are target latency budgets for the prototype. Actual latency may vary depending on the selected providers, network conditions, model, and Vapi configuration.

Figure 1: Kapture Finance AI Collections Voicebot — High-Level Architecture Diagram



3. Conversation Flow and State Machine

3.1 Conversation States

Maya follows a state-based conversation flow. Each state has a defined purpose and allowed actions. The agent cannot skip mandatory states, especially customer authentication.

State	Purpose	Allowed Actions	Exit Condition
CALL_START	Start the outbound call	Initialize call and customer context	Call connected
INTRODUCTION	Introduce Maya and Kapture Finance	State identity and purpose at a high level	Customer engages
CUSTOMER_IDENTIFICATION	Confirm that the intended customer is speaking	Ask for customer name or approved verification information	Customer identified or wrong person
AUTHENTICATION	Verify customer identity	Call <code>verify_customer</code>	Verification successful or failed
ACCOUNT_DISCUSSION	Discuss the overdue account	Reveal overdue information only after successful authentication	Customer understands the situation
INTENT_CAPTURE	Understand the customer's response	Identify will-pay, hardship, dispute, already-paid, callback, etc.	Intent identified
RESOLUTION	Take the appropriate action	Record PTP, send payment link, escalate, or schedule callback	Resolution completed

DISPOSITION	Record the final outcome	Call <code>mark_disposition</code>	Disposition successfully logged
CALL_END	Close the conversation	Give an appropriate closing statement and end the call	Call terminated

3.2 Authentication Gate

Authentication is a mandatory state gate.

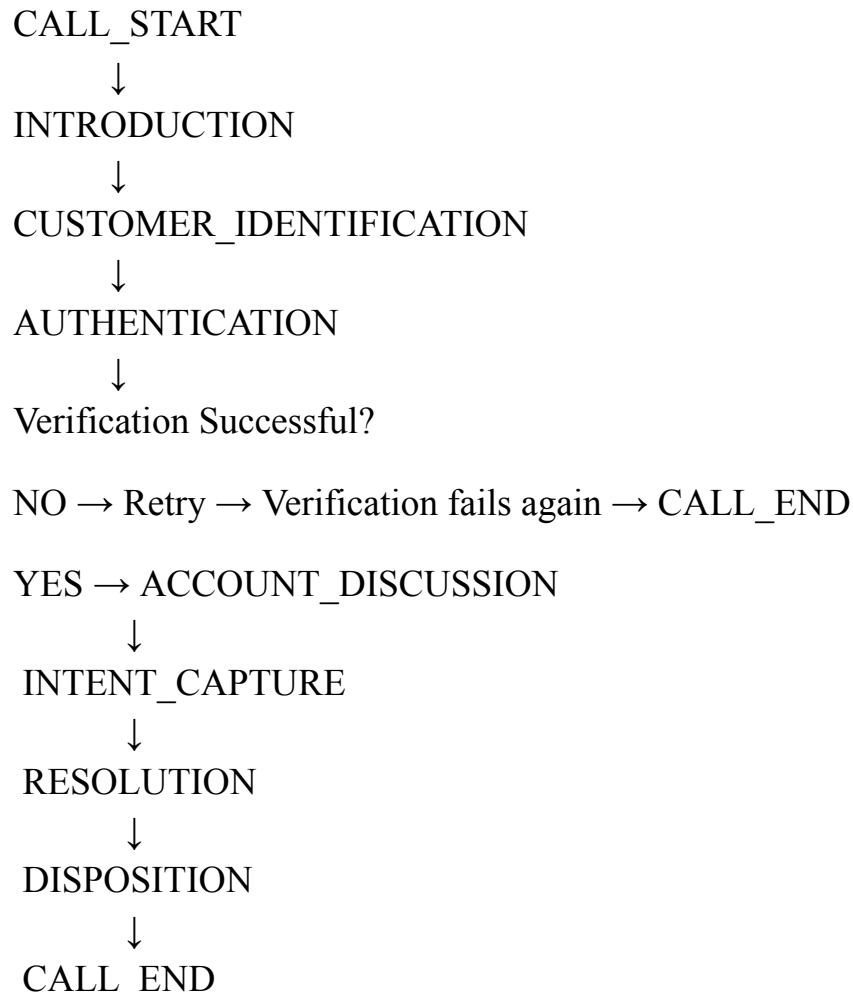
Maya must not enter the `ACCOUNT_DISCUSSION` state unless the `verify_customer` tool returns a successful verification result.

Before successful verification, Maya must not disclose the overdue EMI amount, loan details, payment status, account balance, or any other account-specific information.

If verification fails, Maya should retry according to the configured retry limit. If verification cannot be completed, Maya should end the call or escalate according to the configured business rule.

The authentication requirement must be enforced by the conversation state and backend tool response, not only by the LLM prompt.

3.3 State Transition Flow



3.4 Intent-Based Routing

After authentication and account discussion, Maya identifies the customer's intent and routes the conversation accordingly.

- WILL_PAY → Capture PTP amount and date → **log_promise_to_pay**
- CANNOT_PAY → Understand hardship → Escalate or follow approved resolution
- DISPUTE → Record dispute → **escalate_to_agent**
- ALREADY_PAID → Acknowledge → Route for payment verification
- CALLBACK_REQUEST → Capture preferred time → Schedule callback

- DO_NOT_CALL → Record opt-out → End call
- WRONG_PERSON → Do not disclose information → End call
- WRONG_NUMBER → Record wrong number → End call
- HOSTILE → Remain calm → Escalate or end call
- NO_INPUT → Retry → End call after configured attempts

3.5 Successful Promise-to-Pay Path

1. Maya introduces herself and Kapture Finance.
2. Maya confirms the intended customer.
3. Maya authenticates the customer using `verify_customer`.
4. After successful verification, Maya discloses the overdue EMI information.
5. The customer agrees to pay.
6. Maya asks for the payment amount and expected payment date.
7. Maya calls `log_promise_to_pay`.
8. If requested, Maya calls `send_payment_link`.
9. Maya confirms the result only after receiving a successful tool response.
10. Maya records the final outcome using `mark_disposition`.
11. Maya politely ends the call.

3.6 Already-Paid Path

If the authenticated customer states that the EMI has already been paid, Maya should acknowledge the information without arguing or pressuring the customer.

Maya records the outcome as `ALREADY_PAID` and escalates or routes the case for payment verification. Maya must not claim that the payment has been received unless the backend system confirms it.

3.7 Do-Not-Call Path

If the customer requests that they not be contacted again, Maya should acknowledge the request, record a **DO_NOT_CALL** disposition, and end the call according to the configured compliance process.

4. Intents and Entities

4.1 Customer Intents

Intent	Description	Bot Action
WILL_PAY	Customer agrees to make the payment	Capture amount and payment date, then log PTP
CANNOT_PAY	Customer is unable to pay currently	Understand hardship and escalate appropriately
DISPUTE	Customer dispute the amount or account	Do not argue; record dispute and escalate
ALREADY_PAID	Customer states that the EMI has already been paid	Acknowledge and route for payment verification
WRONG_PERSON	Someone other than the intended customer answers	Do not disclose account information; end the call
WRONG_NUMBER	Number does not belong to the customer	Record wrong number and end the call
CALLBACK_REQUEST	Customer requests a callback	Capture preferred callback time and record the request
DO_NOT_CALL	Customer requests no further calls	Record opt-out and end the call
HOSTILE	Customer becomes abusive or threatening	Remain calm and escalate or end the call
NO_INPUT	Customer does not respond	Retry a limited number of times, then end the call

4.2 Entities to Extract

Entity	Example
Customer Name	Rahul Sharma
PTP Amount	₹8,499
PTP Date	20 August 2026
Callback Date	15 August 2026
Callback Time	10:30 AM
Dispute Reason	Amount already paid
Preferred Language	English / Hindi

4.3 Data Extraction Rules

Maya should extract only the information required to complete the current conversation. The agent should not guess, invent, or modify customer information.

For a Promise-to-Pay (PTP), Maya should capture the committed amount and payment date. For a callback request, Maya should capture the preferred callback date and time. For disputes, Maya should capture the reason for the dispute.

Account-specific information must always come from an authorized backend tool or API response. If required information is unavailable, Maya should ask the customer for clarification or escalate the case rather than making assumptions.

5. Tools / API Design

Maya uses backend tools to perform actions that require trusted system data or a persistent business operation. The LLM should not assume that an action has been completed. It must call the appropriate tool and rely on the tool response before confirming the result to the customer.

5.1 Tool Overview

Tool	Purpose	Key Inputs	Output
verify_customer	Verify customer identity before account disclosure	Customer ID, name, verification information	Verification status
log_promise_to_pay	Record customer's payment commitment	Customer ID, amount, payment date	PTP status and reference ID
send_payment_link	Generate/send payment link	Customer ID, amount, channel	Payment link and delivery status
escalate_to_agent	Transfer cases requiring human assistance	Customer ID, reason, conversation summary	Escalation status
mark_disposition	Record final call outcome	Customer ID, disposition, notes	Disposition status

5.2 Tool Safety Rules

Tools are intentionally scoped to specific business actions.

`verify_customer` must succeed before any account-specific information is disclosed. Payment commitments must be recorded using `log_promise_to_pay`, and Maya must not claim that a payment link was sent unless `send_payment_link` returns a successful response.

Every completed call must end with a disposition recorded through `mark_disposition`. If a tool fails, Maya must not pretend that the requested action succeeded. It should retry where appropriate or escalate the case.

6. Authentication & Data Safety

6.1 Authentication

Maya must verify the customer's identity before discussing any account-specific information. The authentication process is implemented as a mandatory state gate and is not left to the discretion of the LLM.

Maya first confirms that the intended customer is speaking and then calls `verify_customer`. Only when the tool returns a successful verification result can Maya enter the `ACCOUNT_DISCUSSION` state.

Before successful verification, Maya must not disclose the overdue EMI amount, loan details, payment status, account balance, or other sensitive account information.

6.2 Third-Party Protection

If someone other than the intended customer answers the call, Maya must not reveal that the call concerns an overdue loan or disclose any financial information.

Maya should politely ask to speak with the intended customer. If the intended customer is unavailable, Maya should end the call without revealing the purpose of the call.

6.3 Data Safety

The system should follow data-minimization principles and expose only the information required for the current conversation. Sensitive customer data should be retrieved through authorized backend tools rather than being generated or assumed by the LLM.

Tool responses should be validated before being communicated to the customer. Call logs should contain the required operational information while avoiding unnecessary sensitive data.

6.4 Authentication Failure

If customer verification fails, Maya should not disclose account information. The agent may retry verification according to the configured retry limit. If verification continues to fail, Maya should end the call or escalate according to the business rules.

7. Guardrails & Compliance

7.1 Mandatory Disclosures

Maya must clearly identify herself and Kapture Finance at the beginning of the conversation. She should state the general purpose of the call without revealing account-specific information before customer authentication.

7.2 Collection Conduct

Maya must maintain a polite, respectful, and non-threatening tone throughout the call. The agent must not use threats, harassment, intimidation, misleading statements, or pressure tactics.

Maya should provide factual information obtained from authorized backend systems and must not invent payment amounts, dates, account details, policies, or consequences.

7.3 Calling Hours

Outbound calls should be restricted to the permitted calling hours configured by the lending client and applicable regulations. Calls outside the permitted window should not be initiated.

7.4 Do-Not-Call and Opt-Out

If the customer requests no further calls, Maya must acknowledge the request, record a **DO_NOT_CALL** disposition, and stop the collections conversation. The opt-out request should be persisted so that future outbound calls can be prevented according to the client's policy.

7.5 Hallucination and Off-Topic Guardrails

Maya must only answer questions related to the collections conversation and approved customer-service topics. If the customer asks for information outside the agent's scope, Maya

should politely state that she cannot assist with that request and offer escalation where appropriate.

Maya must never invent information. When the required information is unavailable, she should use an authorized tool, ask for clarification, or escalate the case.

7.6 Sensitive Information Protection

Maya must not disclose customer financial information to an unverified person. Account-specific information can only be discussed after successful authentication through the `verify_customer` tool.

7.7 Language Handling

Maya should support English and Hindi. If the customer switches language during the conversation, Maya should respond in the customer's preferred language where supported while maintaining the same authentication, compliance, and conversation-state rules.

7.8 Guardrail Summary

Risk	Guardrail
Debt disclosed to wrong person	Authentication gate
Customer requests no calls	<code>DO_NOT_CALL</code> disposition
Incorrect information	Backend tool verification
Aggressive collection behavior	Polite, non-threatening communication
LLM hallucination	Restricted scope and tool-based data
Out-of-hours calls	Calling-hour validation
Abusive customer	Calm response and escalation
Language switch	English/Hindi handling

8. Edge Cases & Exception Handling

8.1 Edge-Case Handling

Scenario	Expected Bot Behaviour
Already Paid	Acknowledge the customer's statement, do not argue, record ALREADY_PAID , and route for payment verification.
Dispute	Listen to the customer's concern, record the dispute reason, avoid arguing, and escalate to a human agent.
Wrong Person	Do not disclose the purpose of the call or any account information. Politely ask for the intended customer and end the call if unavailable.
Wrong Number	Do not disclose account information. Record WRONG_NUMBER and end the call.
Do-Not-Call Request	Acknowledge the request, record DO_NOT_CALL , and end the collections conversation.
Callback Request	Capture the preferred callback date/time and record the callback request.
Customer Cannot Pay	Listen to the customer's situation, capture relevant hardship information, and escalate or follow the approved resolution process.
Hostile Customer	Remain calm and professional. Do not argue or threaten. Escalate or end the call when appropriate.
No Input / Silence	Prompt the customer again a limited number of times. If there is still no

	response, record NO_RESPONSE and end the call.
Language Switch	Switch between English and Hindi when supported while maintaining the same authentication and compliance rules.
Tool/API Failure	Do not claim that the action succeeded. Retry where appropriate or escalate the case.
Voicemail	Do not disclose debt or sensitive account information. Follow the configured voicemail policy and record the appropriate disposition.

8.2 Failure Handling

The agent should fail safely when an unexpected situation occurs. If a backend tool is unavailable, Maya must not invent a result or tell the customer that an action was completed. The agent should retry within configured limits or escalate the case to a human agent.

Every call, including unsuccessful or interrupted calls, should receive an appropriate disposition whenever technically possible.

8.3 Priority of Rules

When multiple conditions occur during a call, safety and compliance rules take priority over collections objectives. Authentication, privacy, do-not-call requests, and escalation rules must take precedence over attempting to obtain a payment commitment.

9. Escalation & Disposition

9.1 Human Escalation Rules

Maya should handle routine collections conversations autonomously. A human agent should be involved when the conversation requires human judgment, exception handling, or access to information outside the bot's approved scope.

Escalation should occur for disputes, hardship cases requiring special assistance, repeated authentication failures, customer complaints, abusive or highly sensitive interactions, unresolved tool failures, or any situation where the customer explicitly requests a human agent.

Before escalation, Maya should provide the human agent with a concise conversation summary and the relevant disposition information available from authorized systems.

9.2 Escalation Scenarios

Scenario	Action	Disposition
Customer dispute the debt	Escalate to human agent	DISPUTE
Customer reports financial hardship	Escalate according to hardship process	CANNOT_PAY
Customer requests a human	Transfer to human agent	HUMAN_REQUEST
Repeated authentication failure	End or escalate according to policy	AUTH_FAILED
Backend/tool failure	Retry or escalate	TECHNICAL_FAILURE
Customer becomes abusive	Remain calm and escalate/end call	HOSTILE
Customer says EMI is already paid	Route for payment verification	ALREADY_PAID

9.3 Call Disposition

Every call must end with a recorded disposition. The disposition provides a structured summary of the outcome and enables follow-up actions, reporting, and performance analysis.

Maya should call `mark_disposition` before ending the call whenever the backend is available.

Use these standard dispositions:

- PROMISE_TO_PAY
- CANNOT_PAY
- DISPUTE
- ALREADY_PAID
- WRONG_PERSON
- WRONG_NUMBER
- CALLBACK_REQUEST
- DO_NOT_CALL
- HUMAN_REQUEST
- AUTH_FAILED
- TECHNICAL_FAILURE
- NO_RESPONSE
- ESCALATED

9.4 Call Closure

Maya should close every conversation politely and clearly. The closing message should confirm the action taken, if applicable, without making unsupported claims. After the final disposition is recorded, Maya should end the call.

10. Observability & Metrics

10.1 Logging

The system should maintain structured logs for every call to support debugging, compliance review, and continuous improvement. Logs should capture operational information without unnecessarily storing sensitive customer data.

Category	Information Logged
Call Information	Call ID, timestamp, call duration, call status
Conversation State	Current state and state transitions
Authentication	Verification success/failure and retry count
Intent	Detected customer intent
Tool Usage	Tool called, execution status, latency, and errors
Disposition	Final call outcome
Escalation	Escalation reason and outcome
Language	Language used during the conversation
Errors	STT, LLM, TTs, telephony, and backend errors

10.2 Key Metrics

The following metrics should be monitored to measure agent performance, reliability, and business effectiveness.

Metric	Purpose
Containment Rate	Percentage of calls completed without human intervention
PTP Rate	Percentage of eligible calls resulting in a Promise-to-Pay
Average Call Duration	Measure Conversation Efficiency
Average Response Latency	Measure how quickly the bot responds
Tool Success Rate	Measure reliability of backend tools
Authentication Success Rate	Measure successful customer verification
Call Drop Rate	Measure calls ending unexpectedly
Escalation Rate	Measure how often human intervention is required
Do-Not-Call Rate	Tracks opt-out request
Intent Accuracy	Measures whether customer intent is correctly identified

10.3 Latency Budget

The voice experience should remain responsive enough to feel natural. The following initial latency targets can be used for the prototype and refined during testing.

Component	Target Latency
Telephony / Network	≤ 300 ms
STT	≤ 500 ms
LLm / Orchestration	≤ 1000 ms
Tool/API Calls	≤ 500 ms
TTS	≤ 500 ms
Target End-to-End Response	$\leq 2\text{-}3$ seconds

10.4 Monitoring and Debugging

When a call fails or behaves unexpectedly, the team should trace the call using the call ID and inspect the state transitions, model responses, tool calls, tool errors, and latency information. This allows issues to be isolated to telephony, STT, LLM reasoning, tool execution, or TTS.